

Apply for a Science Corps Fellowship - Entry #3220

Name

Alaa Farghli

Emailaf547@cornell.edu**Gender**

Male

City, state, and country of residence

Ithaca, New York, USA

Nationality

Egyptian-American dual citizen

When did or will you finish your PhD?

November 2023

University Education

Ph.D. | Genetics | Cornell University | November 2023

B.S. | Biochemistry | State University of New York at Potsdam | May 2017

What are your reasons for applying to be a Science Corps fellow?

“For a better life,” my father finally shared his reason for immigrating to America while the 2011 Egyptian Revolution unfolded. I was too young to understand the gravity of his words until a family trip to Egypt when I was 19. My cousin, Mohammed, shared his lack of educational and career opportunities caused by Egypt’s crumbling education system, and I observed the cycle of poverty that entraps my family abroad.

As a first-generation student of immigrant parents pursuing my Ph.D. at Cornell University, I had little guidance from my family on pursuing my

educational or professional goals. I am acutely aware that I have been afforded opportunities for growth, learning, and leadership that many others do not have access to. Such chances do not arise in a vacuum. They must be actively created, like the programs under Science Corps. As a Science Corps Fellow, I will continue to build my science communication and teaching skills while uniquely positioned to work with students directly over a long period. This opportunity will let me act as a catalyst to expand the accessibility of education to the next generation of scientists, as other mentors have done for me.

In addition to your studies, describe any relevant experiences that have prepared you for a Science Corps fellowship.

I have dedicated my time to academic and community-building endeavors throughout my academic journey. As an undergraduate at the State University of New York at Potsdam, I served as a teaching assistant for organic chemistry and volunteered as a tutor for first-generation and low-income students. This experience sparked my passion for science education and, as a graduate student, led me to exceed my program's required teaching assistantship obligations by three semesters.

At Cornell, I participated in the Graduate Student School Outreach Program to prepare my own science lessons for local elementary school students. I also conducted a coding workshop for the Expanding Your Horizons program, which introduces middle school girls to science. These programs showed me the importance of making science engaging early to help students stay enthusiastic about studying science. Additionally, I founded and am currently the president of the Arab Graduate Student Association, which facilitates events between Arab students and faculty on campus. In just 1.5 years, it now has over 25 members.

These experiences have prepared me to be successful as a Science Fellow, and I am committed to promoting these values and making science accessible and engaging to all.

Where do you imagine yourself professionally in the future? Check all that apply.

Academic scientist in academia

Administrative role in industry or academia

In science education

In science outreach

Expand on your selection above and describe where you imagine yourself professionally in the near (two year) and long-term (ten year) future.

I will prioritize mentoring opportunities throughout my professional journey regardless of my career path. Thus, I am drawn to the close-knit environments of academic labs. Although my initial training was in biochemistry and molecular biology, during the height of the pandemic, I honed my skillset and passion for bioinformatics. My ultimate goal is to intersect my appreciation for bioinformatics with my demonstrated aptitude for teaching and mentoring.

In the short term, I plan to pursue a postdoctoral position that will allow me to continue developing my bioinformatics expertise and mentor new graduate students. In the long run, my goal is to work within and eventually lead a bioinformatics core. This would allow me to organize workshops that cover the most used computational genomics tools, which early or late-career academics can attend to learn how to analyze their data. As a Science Corps fellow, I look forward to the challenge of teaching and mentoring fledgling scientists. The skills I gain and continue to shape during this period, from lesson planning and the flexibility required when teaching, will be instrumental in developing my pedagogy in the near and long-term future.

CV/Resume

 [ARFarghli_CV.pdf](#)

How did you find out about Science Corps?

University email list

Any other comments/questions?

I will tentatively defend my thesis in early November and receive my degree by the end of 2023. If I am selected as a Science Corps fellow, I am available to start in 2024 and request to be placed in the American School of Tangier in Morocco. In most settings, there is a disconnect between the backgrounds of mentors and mentees. Scientific advancement has been, and continues to be, a worldwide effort transcending background, language, color, and status. However, it is important for students to work with and learn from mentors with similar backgrounds. As an Egyptian-American, I could most closely connect with students in Morocco due to our similar cultural backgrounds.

[Science Corps](#)